

Reference excerpt 01-017

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represented in Fig. 1 as a tube with a circular section. G indicates a reference point on the right IJV. The $w(z, r, t)$ function represents the velocity of the blood in the z direction. It depends on z coordinate, on r (the distance from the axis z) and on t (the time). The symbol $w(z, t)$ is used to indicate the blood velocity averaged over the CSA. The pressure at the right end of the tube ($z = 0$) is the residual arterial pressure and it is assumed to have a constant value p_r while at the left end of the tube ($z = z_{RA}$) the pressure feels the effects of the RA activity varying periodically with the cardiac cycle. The pressure at $z = z_{RA}$ (see Fig. 1) is supposed to be the sum of a component p_c constant in time and of a periodic component $p_a(t)$ which varies according to the atrial cardiac activity. The blood flow is due to the pressure gradient between the ends of the cylinder. It produces a velocity $w_T(t)$ that results from a constant component w_0 due to the constant gradient $p_c - p_r$ and a time varying component $w(t)$ due to the time varying $p_a(t) - p_r$. This system can be described using the Womersley equation that is a linear differential equation where the unknown is the function w and the pressure gradient ($p(t)/z$) is the source term: $2w(r, z, t) \frac{\partial^2}{\partial r^2} + \frac{\partial}{\partial r} w(r, z, t) - \frac{\partial w(r, z, t)}{\partial t} = \frac{1}{\mu} \frac{\partial p(t)}{\partial z}$